



Isolation, characterization and whole genome analysis of the novel genus *Lederbergvirus*, phage vB_EcoP_E212 infecting enterotoxigenic *Escherichia coli* K88

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ABSTRACT

The newly discovered phage vB_EcoP_E212 (also known as E212) was characterized and its genome was annotated in this study, which was conducted in Jilin, China. Transmission electron microscopy indicates that phage E212 belongs to the class *Caudoviricetes*. This phage exclusively infects enterotoxigenic *E. coli* K88. E212 was found to have a short latent period of 20 min, and a burst size of 125 PFU/cell. Additionally, E212 remained stable at all pH levels (3.0–12.0) and temperatures between -20 and 60 °C. The genome of the phage E212 consists of 38,252 bp dsDNA molecule with a G + C content of 46.98%. The genome is projected to include 53 ORFs but no tRNAs. This phage lacks homologs of virulence factors or antimicrobial resistance genes, but it has lysogeny-related genes. Phage E212 was placed in the genus *Lederbergvirus* as a result of nucleotide sequence alignment and phylogenetic analysis.

1. Introduction

The most frequent type of enteric colibacillosis seen in newborn and early-weaned piglets is diarrheal disease caused by enterotoxigenic *E. coli* (ETEC) (Fairbrother et al., 2005). Adhesins (fimbriae) and enterotoxins (proteins), which are both crucial for the progression of the disease, are the key virulence characteristics of ETEC (Nagy et al., 2005). Among the different ETEC, including K88 (F4), K99 (F5), 987p (F6), F41 and F18, the K88 or F18 are the most prevalent strains involved in infections (Huang et al., 2018). Due to the capacity of fimbrial adhesins to attach to glycoprotein receptors on the enterocyte brush border, *E. coli* can invade the small intestine. Once the tight connection integrity is altered and the paracellular passageways of ions, solutes, and water are disrupted, the bacteria release heat-labile enterotoxins or heat-stable enterotoxins, which cause diarrhea (Wilson et al., 1986). Antibiotics can be administered readily to treat ETEC infection; however, abusing them can have negative consequences, including drug residues in animal products (Mitchell et al., 1998), declining immunity (Lin et al., 2018), and antimicrobial resistance (Gay et al., 2017). Globally, rising antimicrobial resistance poses a serious threat to human, animal, and environmental health. Therefore, it is critical to establish a new strategy as soon as possible to manage, suppress, and eradicate bacterial

infections. Bacteriophages are considered valuable alternative antimicrobial solutions (Carvalho et al., 2017). Bacteriophages are viruses that invade bacterial cells and are the most abundant biological entities on earth. They perform critical functions in preserving the diversity and quantity of microorganisms in their natural habitats (Mathur et al., 2003). Phage therapy has a number of significant advantages over traditional antimicrobial therapy, including high bactericidal activity, low inherent toxicity, low cost, and a limited potential for introducing resistance. In this study, we isolated and characterized a novel bacteriophage from farm sewage. Its biological properties were evaluated, and the genome sequence was completely sequenced and analyzed comparatively with its related phages. These results were used to determine its potential for the prevention and control of enterotoxigenic *E. coli* K88.

2. Materials and methods

2.1. Bacterial strains and phage isolation

Table S1 list the bacterial strains employed in this study. The host strain enterotoxigenic *E. coli* K88 (CVCC 83,902) was obtained from China Veterinary Culture Collection Center (CVCC). All strains were

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grown on LB medium with constant shaking at 37°C or were kept at –80°C in LB medium containing 50% glycerol. Phage vB_EcoP_E212 (also known as E212) was isolated and purified from farm sewage in Jilin province using the methods previously described by Zhang et al. (Zhang et al., 2013). Bacteriophage lysates were purified using a sterile disposable membrane filter (0.22 µm) and kept at 4°C.

2.2. Host range investigation and efficiency of plating analysis

Twenty-two bacterial strains, including strains of *E. coli*, *Salmonella* and *Shigella* (Table S1) were utilized to investigate the lytic capacity of phage E212 using the spot test method (Jun et al., 2014) to evaluate its potential to create lysis zones on lawn cultures of various bacterial strains. The purified phage E212 suspension (10 µL, 10⁹ PFU/mL) was spotted directly onto the surface of a bacterial lawn culture plate and incubated at 37°C overnight. The presence of clear zones surrounding the phage drop on the plates was examined. To assess the host spectra of the phage against a variety of bacterial strains, efficiency of plating (EOP) was computed based on phage titer on the test strains versus the phage titer on the host bacteria (positive spot test). The result for the host strain was taken to be EOP=1.

2.3. Transmission electron microscopy

AJEM-2100EX transmission electron microscope (TEM) was used to examine the purified phage suspension (10⁹ PFU/mL) after incubation for 10 min on copper grids covered with carbon (JEOL Co., Tokyo, Japan) (Yuksel et al., 2001).

2.4. Chloroform tolerance

Purified phage E212 suspension (100 µL) was mixed with 900 µL of chloroform and incubated for 1 h at 37 °C. The double-layer agar plate method was used to determine the resulting phage titer.

2.5. Heat and pH stability

The heat stability of E212 was tested at –20, 4, 25, 40, 50, 60, 70 and 80°C in LB broth, after incubation for 0, 40 and 60 min. To test the pH stability, E212 was suspended in LB broth at different pH values (adjusted from 1 to 13 with HCl and NaOH) and incubated for 2 h at 37°C. The remaining phage particles were quickly diluted after incubation. The double-layer agar plate method was used to determine the titers of the test phage.

2.6. Multiplicity of infection

The proportion of virus particles to host cells is known as the multiplicity of infection (MOI). In a spectrophotometer (L9, Shanghai INESA Analytical Instrument Co., Ltd, Shanghai, China), the host strain enterotoxigenic *E. coli* K88 (CVCC 83,902) was grown at 37 °C to an absorbance at 600 nm (OD₆₀₀=0.803), which corresponded to a cell count of approximately 1 × 10⁸ CFU/mL. According to the MOIs, the ratios between E212 and the host strain were 0.0001, 0.001, 0.01, 0.1, 1 and 10. The phage lysate was centrifuged at 12,000 rpm for 7 min after being incubated at 37 °C for 4 h. To calculate the phage titer, the supernatant was diluted. The ideal MOI was thought to be the one with the highest titer.

2.7. One step growth curve

The following modifications were made to the one step growth curve described earlier by Yuan et al. (Yuan et al., 2019), Cong et al. (Cong et al., 2021). The host strain was grown to a mid-exponential phase (OD₆₀₀=0.803), harvested by centrifugation (8000 × g for 5 min at 4°C) and resuspended in LB broth to an adjust density of ~10⁸ CFU/mL.

Phage E212 was added at an optimal MOI of 0.001, and incubated at 37°C for 10 min, and no adsorbed phages particles were moved. Bacterial pellets were suspended in 10 mL of LB medium, followed by incubation at 37 °C with orbital shaking (160 rpm). The double-layer agar plate method was used to plate samples for phage titer after they had been collected at intervals of 10 min over a period of two hours. By subtracting the initial titer from the final titer and then dividing the result by the original titer, the burst size was estimated (Hamdi et al., 2017). At the midway of the exponential phase of the curve, the latent period was computed (Duplessis et al., 2005).

2.8. Phage infection experiments in vitro

With the exception of the following changes, phage infection studies were carried out as previously described by Raya et al. (Raya et al., 2014). Phage E212 was introduced to the host bacterium in the early exponential phase (OD₆₀₀=0.803, CFU/mL) and quickly mixed. After 30 min of sample treatment with 100 µL of chloroform, the phage titer (PFU/mL) was calculated, and the number of surviving host bacterial cells was counted (CFU/mL). The phage titers were measured using the conventional double layer approach in triplicate. On LB plates, bacterial survival was also measured in triplicate.

2.9. Phage DNA purification and sequencing

According to Sambrook et al. (Sambrook et al., 2001), phenol-chloroform-isoamyl alcohol was used to extract phage genomic DNA from a preparation containing a high concentration of phage particles (10¹⁰ PFU/mL). A DNA library was created in accordance with the Illumina TruSeq™ Nano DNA Sample Prep Kit instruction. Utilizing the Illumina NovaSeq sequencing platform (150 bp×2) and paired-end reads, Shanghai Personalbio Technology Co., Ltd. (China) carried out whole-genome sequencing (Bolger et al., 2014). The sequence data collected amounted to 382 Mb in total. A 150 bp read was considered to be the average. Trimmomatic v 0.36 was used to filter out low-quality reads (Q-value<20, 98.51%) from the total of 25,525,764 reads, with a depth of coverage of approximately 98,722. To construct high-quality sequencing data to constrain contigs and scaffolds, A5-MiSeq v20160825 (<https://arxiv.org/abs/1401.5130>) and SPAdes v3.12.0 (<http://cab.spbu.ru/files/release3.12.0/manual.html>) were employed. Next, the remaining inner local gaps were filled, and the single-base polymorphism was fixed for the final assembly results using MUMmer v3.1 (<http://mummer.sourceforge.net/>) and Pilon v1.18 (<https://github.com/broadinstitute/pilon>).

2.10. Genome analysis

Open reading frames (ORFs) were identified using the RAST server (<http://rast.nmpdr.org/rast.cgi>) and verified GeneMark Server (<http://topaz.gatech.edu/GeneMark/genemarks.cgi>). The current protein and nucleotide databases (<http://www.ncbi.nlm.nih.gov/>) were searched using the final assembled genome sequence using the Basic Local Alignment Search Tool (BLAST). The putative functions of the encoded proteins were determined using BLASTp (<https://blast.ncbi.nlm.nih.gov/Blast>), and the homologous proteins were searched using hmmer (<https://toolkit.tuebingen.mpg.de/tools/hmmer>). The similarity of phage genome sequences was assessed using BLASTn (<https://blast.ncbi.nlm.nih.gov/Blast>).

Utilizing tRNAscan-SE, putative tRNA sequences were located (Schattner et al., 2005). The ResFinder sever (Zankari et al., 2012) (<https://cge.cbs.dtu.dk/services/ResFinder/>) and Virulence Factor Predictor (Joensen et al., 2014) (<https://cge.cbs.dtu.dk/services/VirulenceFinder/>) were used to identify antimicrobial resistance determinates in the E212 genome.

A phylogenetic study of the whole genome sequence of phage E212 using 6 known sequences based on the latest International Committee on

Taxonomy of Viruses (ICTV) Taxonomy Release (belonging to the genus *Lederbergvirus*) and 12 known sequences in GenBank was carried out by MEGA-X. After ClustalW sequence alignment, the maximum likelihood approach was used to construct the phylogenetic tree. The large subunit terminase sequence was used to ascertain the phage's DNA packaging methods. The ClustalW algorithm was used to select the large subunit terminase amino acid sequence from this study and those from other phages for multiple alignments, and MEGA7's neighbor-joining approach was used to build a phylogenetic tree. Using Easyfig, comparative analyses of the full genome sequences of the phage E212 and the other *Lederbergvirus* species were performed.

2.11. Statistical analysis

For each time point, averages and standard deviations were computed across all experiments, which were run in triplicate. To examine differences between the two groups of samples, an unpaired t-test was used. *P* values less than 0.05 were regarded as statistically significant for all analyses performed using GraphPad Prism 7.0.0 (Graph Pad, CA, USA).

3. Results

3.1. Bacteriophage isolation, morphology and host range of phage

From sewage, one phage directed against enterotoxigenic *E. coli* K88 (CVCC 83,902) was identified. This phage's name is vB_EcoP_E212 (referred to as E212). According to TEM research, E212 features an icosahedral head (50 ± 3 nm) attached to a tail (8 ± 2 nm). E212 was classified as a member of the *Caudoviricetes* class based on these structural characteristics (Fig. 1). The infectivity of E212 was evaluated on 22 bacterial strains (*E. coli* *n* = 14; *Salmonella* spp. *n* = 4; *Shigella* spp. *n* = 4) using a spot test to determine its host range (Table S1). Phage E212 had a very narrow host range and inhibited only the host bacteria (CVCC 83,902), resulting in clear plaques.

3.2. Chloroform tolerance and MOI

When exposed to chloroform, phage E212 remained stable (*P* > 0.05) (Fig. 2a). The MOI for the highest titer among all examined MOIs was 0.001 (Fig. 2b).

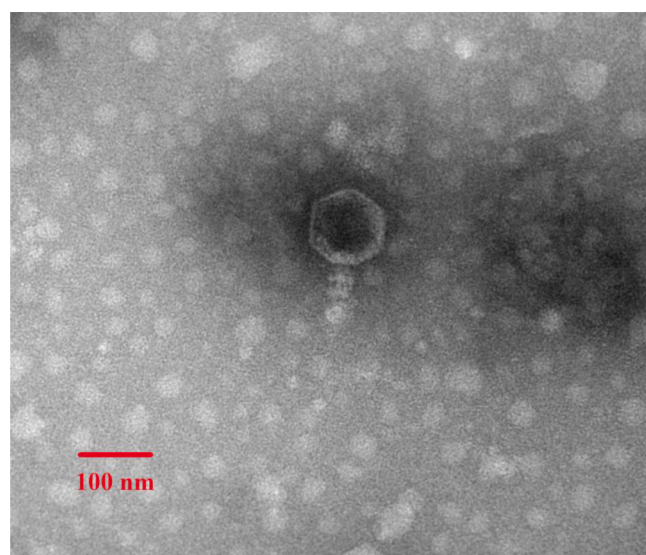


Fig. 1. Transmission electron micrograph of E212 negatively stained with 2% (w/v) uranyl acetate. The scale bar represents 100 nm.

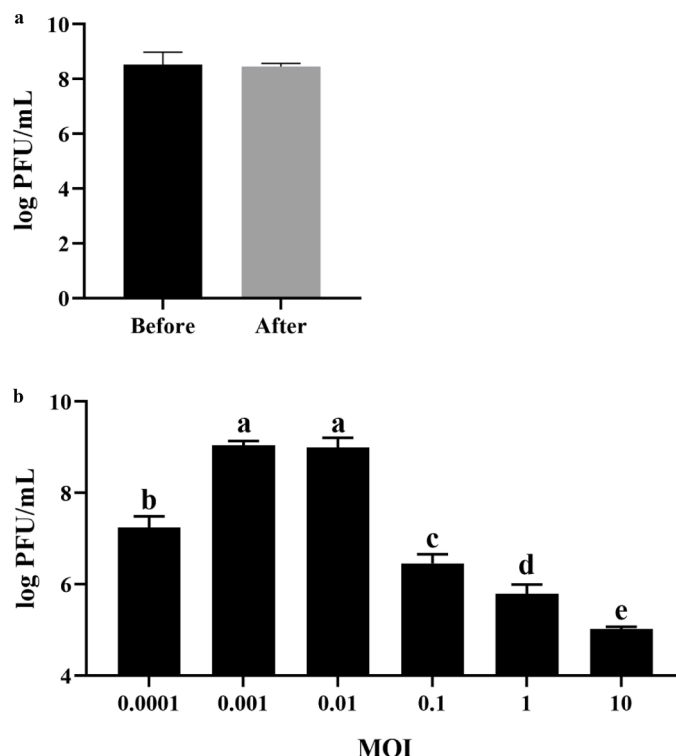


Fig. 2. a Chloroform stability. b Titrers of phage E212 produced at different MOIs.

3.3. Heat and pH stability

Phage E212 activity remained at a high level during a heat stability test at temperatures ranging from -20 °C to 60 °C. However, higher temperatures had an impact on the survivability of phage E212, and the titer of phage E212 incubated at 70 °C for 40 min decreased. When the temperature was increased to 80 °C incubation for 40 min and 80 min, the titer of phage E212 rapidly declined (approximately 70% and 90%, respectively) (Fig. 3a). Phage E212 showed stability between pH 3 and pH 12. After incubation for 1 h at pH < 3 or pH > 12, no phage particles survived (Fig. 3b).

3.4. One step curve and phage infection experiments in vitro

The one-step curve showed that phage E212 had a brief latent period of approximately 20 min (Fig. 4a). Following the latent period, there was an outbreak period, which lasted approximately 60 min, from 20 min to 80 min post-infection, and then shifted into a plateau period. The burst size of E212 was 125 PFU/infect cell (host cell enterotoxigenic *E. coli* K88 titer: 7.5×10^8 CFU/mL). The lytic activity of phage E212 was evaluated by measuring the host bacterial survivors, phage titers and optical density of the liquid medium during the growth of host bacteria at 37 °C and MOI = 0.001. The results showed that phage E212 can reduce the number of enterotoxigenic *E. coli* K88 in 6 h (Fig. 4b). The survivors of host bacteria were 1 log lower, and the titers of phage E212 were 4 logs higher than the initial time point.

3.5. Characterization of the phage E212 genome

The results of the present study showed that phage E212 has a linear double-stranded DNA genome with a length of 38,252 bp and an overall G + C content of 46.98%. By using the RAST service, we were able to locate 53 ORFs and predict 44 potential protein-coding genes in the genome, 31 of which had a functional designation. All ORFs accounted for 86.2% (32,976 bp) of the whole genome. The average ORF length

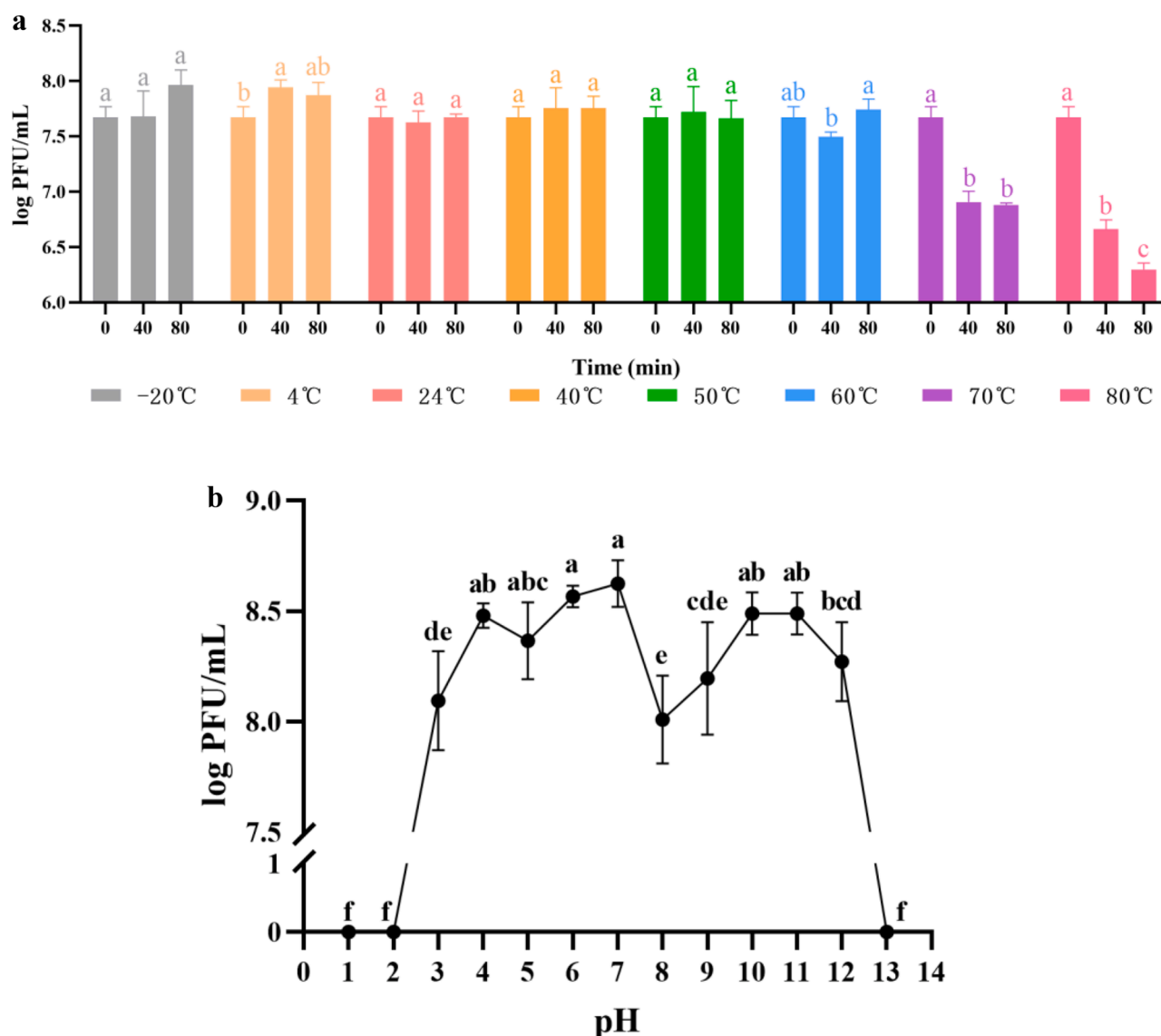


Fig. 3. a The influence of temperature on phage E212 viability. b The influence of pH on phage E212 viability.

was 622.19 bp. No tRNA genes were predicted using tRNAscan-SE. The majority of the ORFs in E212 (47 ORFs, 88.7%), four GUG codons (7.5%), and two UUG codons (3.8%) were expected to start with an AUG codon. The three stop codons were distributed differently; UGA was the most prevalent (31 ORFs, 58.5%), followed by UAA (18 ORFs, 34%) and UAG (4 ORFs, 7.5%).

These ORFs were divided into eleven functional modules based on bioinformatic predictions, including recombination protein, lysogeny control (protein CI and CII), antitermination (protein O, P and Q), Nin genes (Nin B, F and H), host cell lysis (holing, lysin and Rz endopeptidase), phage structure (portal, scaffold, capsid, head assemble protein and injection), DNA packaging (terminase large and small subunit), DNA replication and modification (DNA methylase, DNA binding, DNA transfer and transglycosylase), stabilization proteins, host specificity (tail spike protein), and integration (integrase) (Table S2 and Fig 5).

ORFs 29, 30, and 31 were identified as the predicted portal protein, scaffold protein, and capsid protein, respectively, in the phage structural modules. These proteins come together to create the procapsid. The development of phage scaffold proteins aided in improving bio catalysis for a range of biotechnology applications. The production of phage chaperone proteins and proteases allows only capsids that are homogenous in size and structure to assemble inside the host bacteria (Liu

et al., 2020). ORF 45 was predicted to be a tail spike protein and shared high amino acid sequence similarity with that of Enterobacteria phage HK620 (Identity: 93%, E -value: $9.8E^{-58}$, Accession number: NP_112, 090.1). The tails of a phage are the main parts responsible for host cell attachment and injection. The phage HK620 tail spike, as reported in other investigations, creates hexasaccharides of an O18A1-type O-antigen and exhibits endo-N-acetylglucosaminidase activity (Barbirz et al., 2008). The tail phage terminase enzyme consists of large and small subunits. The DNA motor's N-terminal ATPase and the nuclease's C-terminus, which are in charge of prohead binding, DNA translocation, and DNA cleavage, make up the large subunit (ORF 28) (Cataiano et al., 1995). The small terminase subunit (ORF 27) has DNA binding activity and is involved in the recognition of its own DNA. The headful packaging method used by P22-like phages is made possible by the terminase and portal proteins (Casjens et al., 1992). ORF 46 was predicted to be an integrase, with 99% sequence identity (E -value: $4.3E^{-232}$). Three ORFs were identified as host lysis modules, including the holing module (ORF 21), lysis module (ORF 22) and RZ protein (ORF23).

3.6. Phylogenetic and comparative genomic analysis

The whole genomes of 18 published phages and E212 were subjected

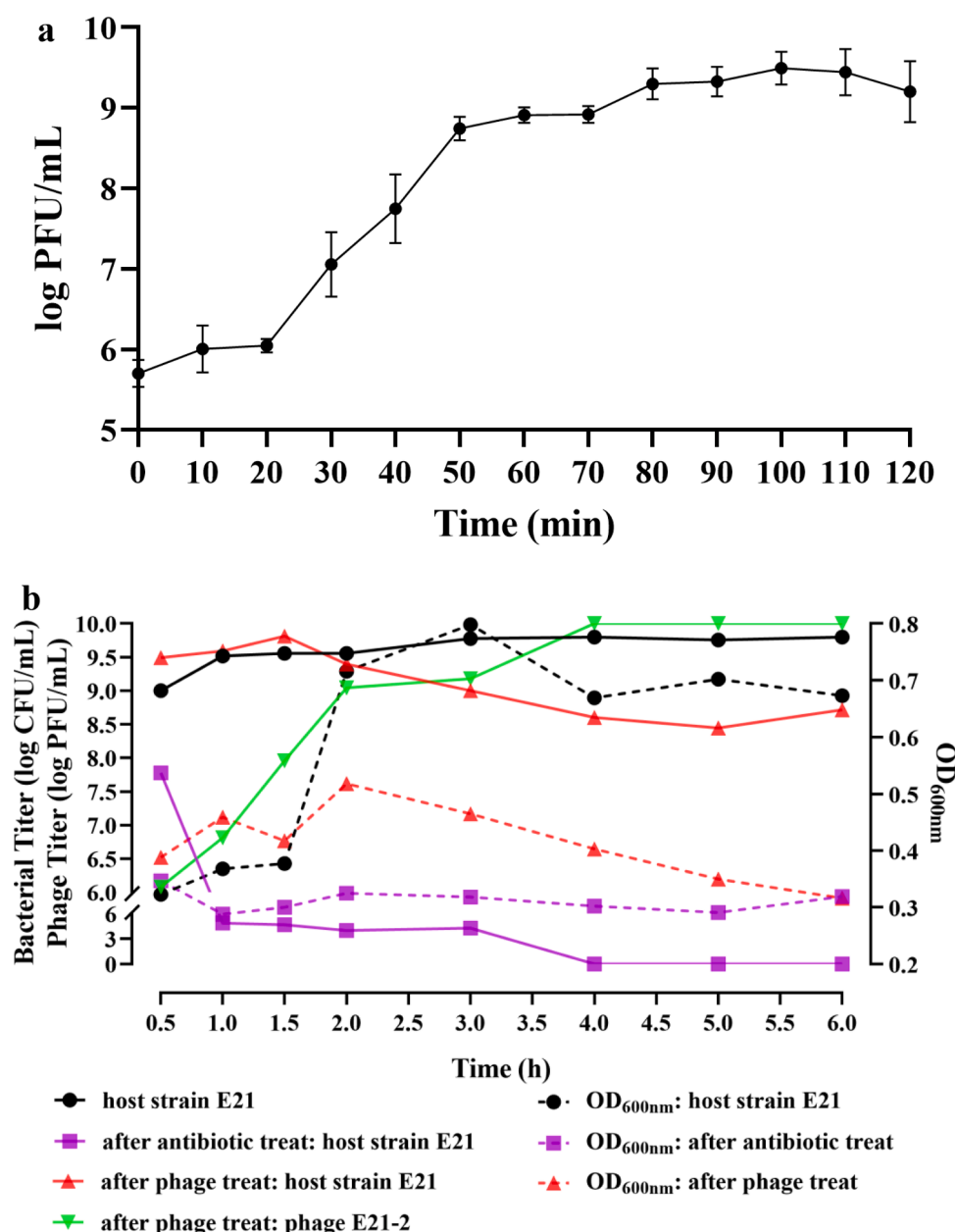


Fig. 4. a One step growth curves of phage E212 in the presence of the Enterotoxigenic *E. coli* K88 host. b Optimal MOI (0.001) co-infection of Enterotoxigenic *E. coli* K88 growing with phage E212. Bacterial survivors, OD_{600 nm} and phage titers.

to phylogenetic analysis to determine the evolutionary relationship between the two phages within the genus *Lederbergvirus* (Fig. 6a). In the phylogram tree, *Escherichia* phage phiv205-1, *Escherichia* phage JEP4, and *Enterobacteria* phage Sf101 were more closely related to phage E212. The DNA packaging methods used by tailed dsDNA phages are divided into six categories and 17 subcategories. These categories are (a) cohesive ends (5' cos, lambda P2; 3' cos, HK97); (b) headful packaging (P2, P22, Sf6, T4, 933 W, phiPLPE, phiKZ); (c) host ends (Mu, D3112); (d) short direct terminal repeats (DTRs) (T7, N4, C-st); (e) long DTRs (SPO1); and (f) covalently bound terminal proteins (*Bacillus subtilis* phage phi29) (Casjens et al., 2009; Merrill et al., 2016). Phage E212 uses the headful packaging method, as shown in Fig. 6b (P22). In the E212 genome, no homologs of virulence factors (*E. coli* virulence genes) or antibiotic resistance genes were discovered. However, lysogenic genes are present in all *Lederbergvirus* phages, including E212. E212 is therefore unsuitable for therapeutic use. While phages are valued alternative antimicrobial remedies, not all phages can be used as therapeutic agents.

Phage genome sequences containing antibiotic resistance, virulence, and lysogenic genes are not allowed due to security concerns.

Among the phages belonging to the genus *Lederbergvirus*, E212 and *Enterobacteria* phage Sf101 have twenty conserved proteins with an identity of 91.87% (Fig. 7). These include the portal, scaffold, capsid, integrase, DNA methylase, DNA binding protein, i-spanin, O protein, holin, and lysis. They also include the head assembly protein, injection protein, tail spike, and capsid protein.

4. Discussion

Due to their abundance and genetic diversity, phages are currently recognized as the most successful biological entities in the world (Rohwer et al., 2015), with an estimated 10^8 virions per milliliter in natural water (Bergh et al., 1989), 10^7 virions per gram in soil (Ashelford et al., 2003), and 10^5 to 10^7 virions per milliliter in sewage (Bitton, 1987). Previous research revealed that sewage from sewage treatment

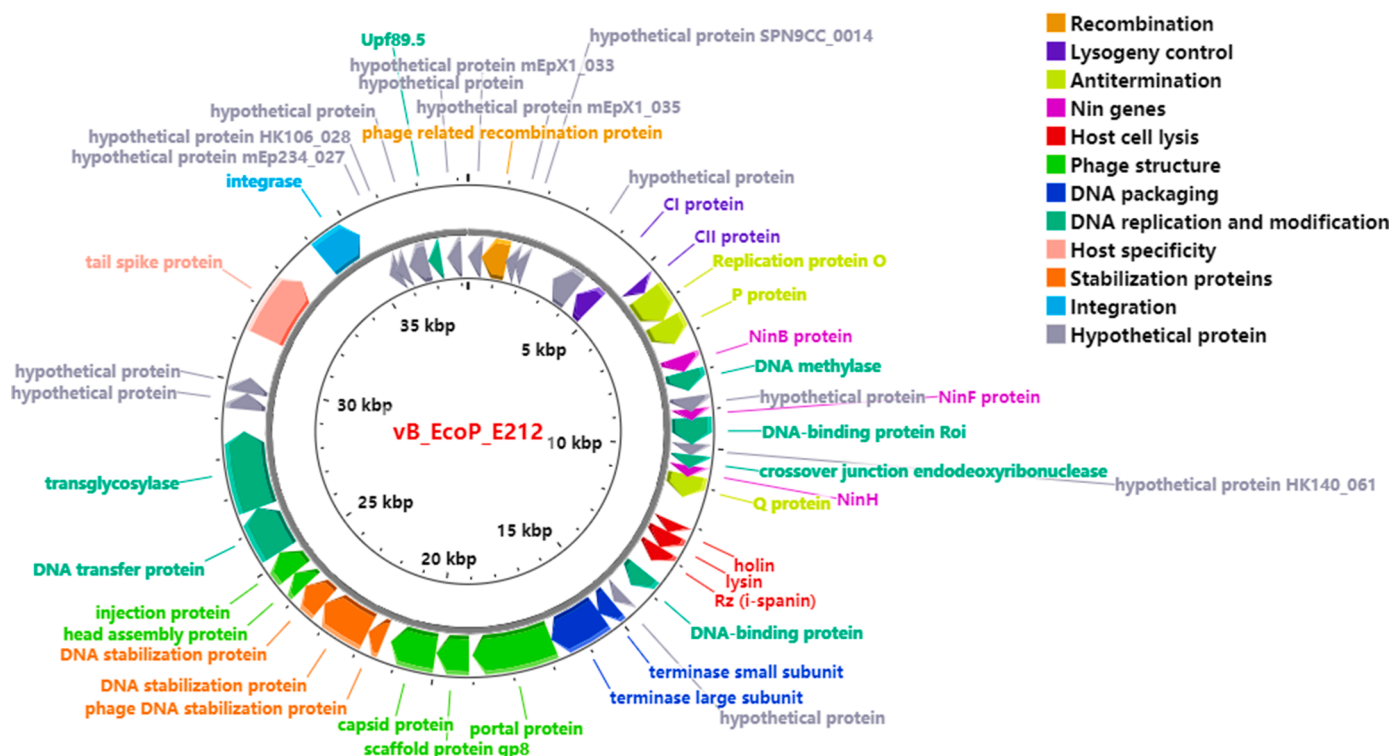


Fig. 5. Predicted functions of phage E212.

plants and animal drainage are the main sources of coliphages, followed by feces and gastrointestinal contents (Osawa et al., 1981). It proved to be very efficient to concentrate phages from farm sewage obtained from farms with a recorded infection with postweaning *E. coli* diarrhea (Jamalludeen et al., 2007). In this study, we isolated a novel enterotoxigenic *E. coli* K88 bacteriophage, vB_EcoP_E212, from pig farm sewage. The phages are devised by the ICTV according to the genome type, phage tail, and capsid/head morphologies. According to the latest ICTV Taxonomy Release, the naming of bacterial viruses has been standardized into the Linnaean nomenclature. Order *Caudovirales* and families *Myoviridae*, *Podoviridae* and *Siphoviridae* were removed, and all underlying taxa were to be assigned directly to the class *Caudoviricetes* (Kropinski, 2018). This class has 4 orders, 47 families, 98 subfamilies, 1197 genera and 3601 species. In our study, morphological features (head 50 ± 3 nm, tail 8 ± 2 nm) showed that phage E212 belongs to the class *Caudoviricetes*. A phage's host range is the variety of bacterial cells it may infect and destroy (Hyman et al., 2010). In general, the host range is defined as adsorptive, penetrative, bactericidal, productive, plaquing, spotting or lysogenic (Hyman, 2010). The spot test can be employed as a simple and rapid method that indicates lysis capacity; however, it does not distinguish between specific and nonspecific lysis and thus cannot precisely signal infection. The effectiveness of the plating (EOP) approach, which quantifies the number of phage offspring produced by a specific phage infecting a bacterial strain, is superior (Hyman, 2010). In our study, E212 had a very narrow host range and could lyse only its host bacteria among 22 tested bacterial strains. This strict host range may be due to the highly specific tail spike protein, which mediated the attachment of the phage to bacterial cell surface polysaccharides (Freiberg et al., 2003). To overcome this limitation, phage cocktails (multiple phages to lyse multiple host bacteria) (Chan et al., 2012) and polyvalent phages (a single phage to kill multiple host bacteria) (Ye et al., 2018) are considered as therapeutic strategies. However, the advantages and disadvantages of monotherapy, polyvalent, or cocktail phage therapy have yet to be investigated.

In general, bacteriophages are less stable in extremely acidic environments ($\text{pH} < 3$) due to protein denaturation (Hazem, 2002). The

optimal pH conditions of most phages range from 5 to 9, but many phages survive at lower (3 to 4) and higher (10 to 11) pH values (Jamal et al., 2015; Jamalludeen et al., 2007). In our experiment, phage E212 showed good stability over a wide pH range from 3 to 12, with maximum stability at $\text{pH} = 7$. Attachment, penetration, multiplication, and the duration of the latent period are all significantly influenced by temperature (Jończyk et al., 2011). Tsutsaeva et al. revealed that survival was 98.4% for phage T3, 80.55% for phage phi X174, and 65.13% for phage T4 after freezing at -196°C (Tsutsaeva et al., 1981). These results agreed with the earlier observation of Kerby et al., who reported that the optimum temperature and long storage stability conditions of phage T7 ranged from 0.5 to 2°C at $\text{pH} 6-8$ (Kerby et al., 1949). In our experiment, phage E212 demonstrated strong resistance to a wide range of temperatures, from -20°C to 60°C .

During phage DNA extraction, purification and concentration, there was considerable bacterial debris, nonlysed bacteria, and some resistant strains, and chloroform was used to preserve phage lysates (Fulton, 1955). To determine the presence or absence of lipid components in the capsids or tail of phages, the chloroform tolerance of phages is a crucial reference point. There was no difference in titer between the untreated phage and the chloroform-treated phage, which indicated that phage E212 did not contain lipids.

In our experiment, the MOI of phage E212 was 0.001. The multiplicity of infection (MOI) concept relates phages to bacteria. Indeed, not all adsorbed phages achieve successful infection (Hyman, 2010). Thus, effective phage adsorption does not necessarily translate into effective phage infection (Abeldon, 2016). The findings of the one-step curve revealed the latent time to be 20 min, the outbreak period to be 30 min, and the burst size to be 125 PFU/infected cell. Grabner et al. (1968) reported that the minimal latent period for *Salmonella* phage MG40 (*Lederbergvirus* genus) was 51 min, while the latent period of P22 was 35 min; both MG40 and P22 phages exhibited similar burst sizes of approximately 100 PFU/infected cell, in agreement with our results. However, Shin et al. (2014) reported a similar latent period of 30 min and a larger burst size of 220 PFU/infected cell for *Salmonella* phage SPN9CC (*Lederbergvirus* genus). These results indicated that the latent

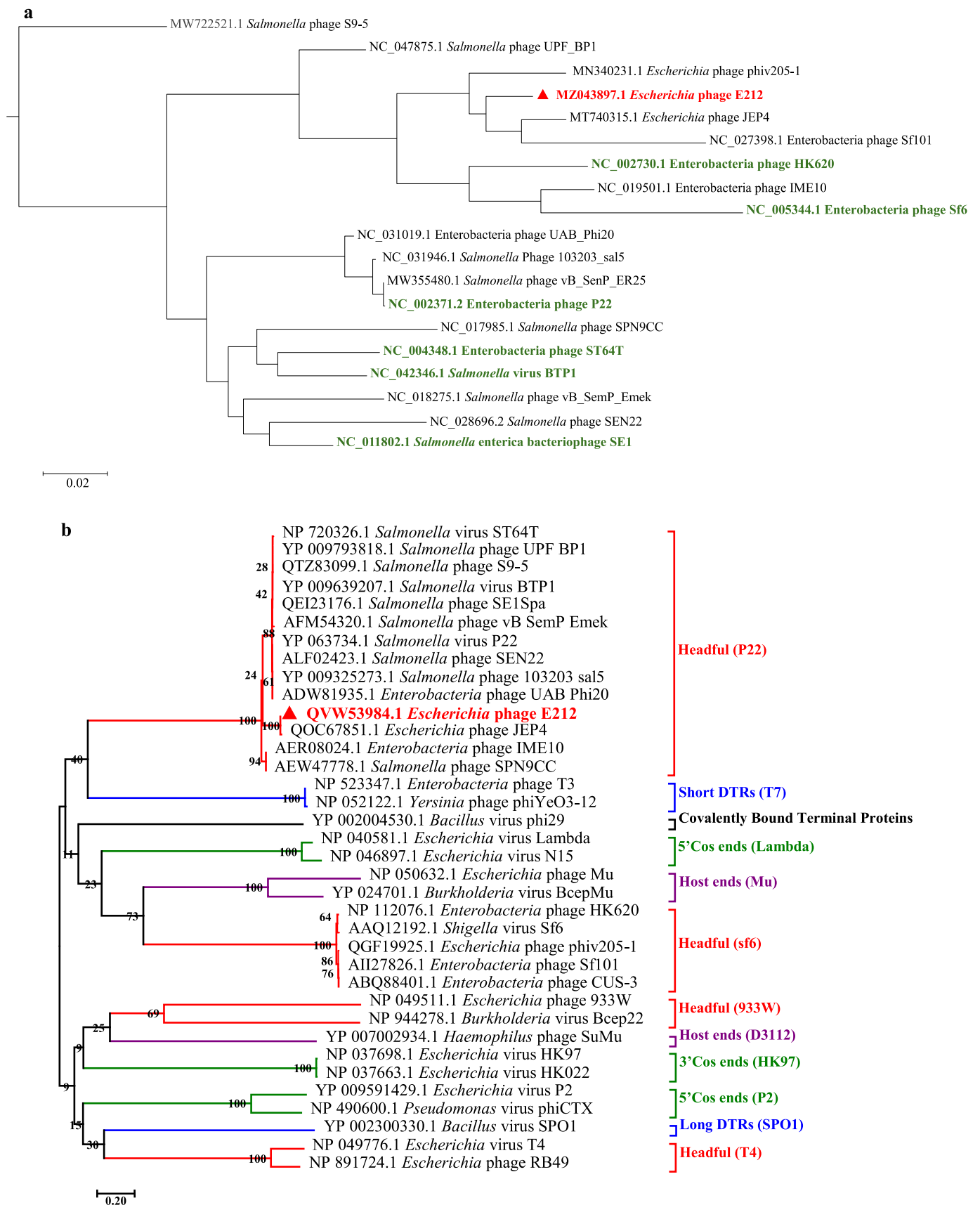


Fig. 6. a The phylogenetic tree of 19 whole genome of phages within genus *Lederbergvirus*. b Neighbor-joining phylogenetic tree based on amino acid sequence of terminase large subunit, showing the relationship between phage E212 and other phages. Values at the nodes indicate the bootstrap support calculated from 1000 replicates.

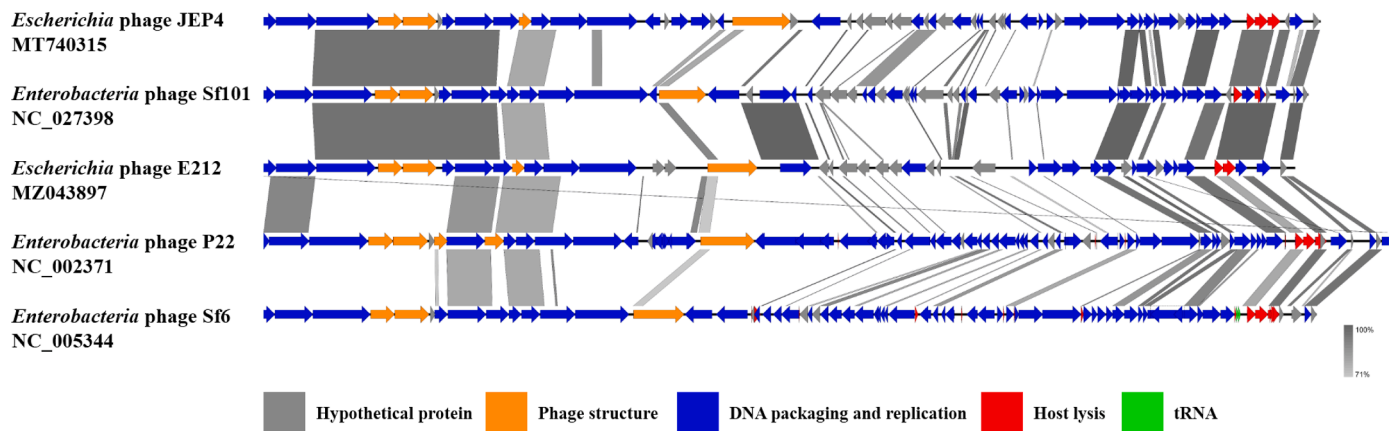


Fig. 7. Comparison of the genome of phage E212 to other *Lederbergvirus* members (*Escherichia* phage JEP4, *Enterobacteria* phage Sf101, *Enterobacteria* phage P22, *Enterobacteria* phage Sf6) using Easyfig software. The different color arrows represent CDS in the whole genome sequence. The direction of arrow indicated the transcription direction of each CDS. The grey bars indicated the similarity of two pairs of sequences, and the intensity of grey indicated the degree of sequence similarity.

period and burst size varied among different phages. Critical to the efficacy of phage therapy are the phage adsorption rate, burst size, latent period, and starting dose (Weld et al., 2004). This study determined the infection ability of host bacteria *in vitro*. Overall, treatment of enterotoxigenic *E. coli* K88 with phage E212 resulted in a significant decrease in the OD₆₀₀ and bacterial titer (PFU/mL) compared to the host strain. The inhibitory effect on strain growth was observed for up to 5 h, and maximal bacterial lysis was observed at the 5-hour time point, but the bacterial titer increased again after 5.5 h. This increase in OD₆₀₀ or bacterial titer may be because of host strain adaptation as bacterial replication continued and phage-resistant bacterial cells survived. It can also be attributed to other factors, such as bacterial debris and metabolic products (Piracha et al., 2014).

The most fundamental method to fully understand the phage structure, evolution, interaction with host bacteria, and safety is to analyze its genome. Thus, an increasing number of phage genomes have been sequenced. Based on typical morphology, and high homology to P22 both at the genome and acid amino level, these phages were defined as members of genus *Lederbergvirus*, including BTP1, HK620, SE1Spa, Sf6, ST64T, among others (<https://ictv.global/taxonomy>; <https://www.ncbi.nlm.nih.gov/genome/?term=Lederbergvirus>). According to analysis genome of the genus *Lederbergvirus*, the genome sizes range from 38.25 kb to 46.07 kb (with an average GC content of 40.5% to 48.7%). Forty-four to eighty predicted proteins were identified in the genome of genus *Lederbergvirus*, 0 to 2 tRNAs were found in all the 59 genomes. In the present study, phage E212 contained a linear double-stranded DNA genome 38,252 bp in length with an overall G + C content of 46.98%. Forty-four predicted proteins and 0 tRNAs were identified in the E212 genome, which showed high genomic homology with genus *Lederbergvirus*. Generally, the genome of a phage can be divided into various functional modules based on their functions: DNA packaging, assembly, DNA modification, replication, and lysis. Phage E212 genes 29, 30 and 31 encode portals, scaffolding and coat protein, respectively, which assemble to create the procapsid. These three functional proteins were required for the procapsid assembly of ds DNA phages, specifically the packaging of the DNA into a preformed protein “container” (Earnshaw et al., 1980). The portal protein, composed of twelve subunits, produces a hole at one icosahedral vertex through which DNA enters and escapes during packaging and ejection, respectively (Casjens et al., 2011). The amino acid sequence of the E212 portal protein was 99% identical to that of *Enterobacteria* phage Sf101 but only 35% identical to *Salmonella* phage P22. In macromolecular assembly, scaffolding proteins play a significantly greater role. Generally, viral scaffolding proteins can be divided into three categories: 1) the icosahedrally ordered, external scaffolds of bacteriophage P4 and ϕ X174-related phage; 2) the internal,

core-like scaffolds, such as those of P22, herpes and λ ; and 3) the prolate core of T4-like phages. Previous research demonstrated that the P22-like scaffolding proteins generated three tree branches with 15% to 27% sequence identity between branches (Casjens et al., 2011). P22-like phage scaffolding proteins are highly divergent, but they are all rich in prolines and charged amino acids (Dokland, 1999). Spectroscopic studies revealed that the isolated subunit’s secondary structure is rich in α -helices (approximately 40%), is extremely thermolabile, and is characterized by *noncooperative* unfolding ($T_m \approx 49^\circ\text{C}$) (Tuma et al., 1996). In this study, the amino acid sequence of the E212 scaffolding protein was 33% identical to that of *Shigella* phage Sf6 but not identical to that of *Salmonella* phage P22. Among the 297 amino acids, 108 (approximately 36.4%) were charged amino acids. Coat protein sequences, such as scaffolding proteins in tail-tailed phages, are exceedingly diverse, and comparing several of them reveals no amino acid commonality. Casjens investigated the variety of coat proteins of the fifty-seven P22-like phages known to exist (Casjens et al., 2011). The results indicated that there are three major sequence types, with pairwise similarities ranging from 14% to 28%. These major types can be split into eight subtypes (P22, Scho1, MS21A, CUS-3, Sf6, ϕ SG1, Rett1 and APSE-1) that differ by >15%; the most divergent pairings of individuals within the P22, CUS-3 and Sf6 groups range from approximately 2% in the P22 branch to approximately 5% in the CUS-3 branch. In this study, the amino acid sequence of the coat protein was 99% identical to *Enterobacteria* phage CUS-3 but only 30% and 27% identical to P22 and Sf6, respectively. Despite their diversity, P22-like phage coat protein structures have been examined as a common ancient ancestor, and those phages have extremely similar shell topology structures (Teschke et al., 2003; Zhang et al., 2000).

The tail complexes of phages play a crucial role in host identification and DNA transport during injection. Prior research demonstrated that the majority of P22-like phage tailspikes contain polysaccharide hydrolase activity that cleaves their O-antigen polysaccharide receptors (Barbirz et al., 2008; Freiberg et al., 2003; Iwashita et al., 1973; Zayas et al., 2007). Variability in the tailspike protein suggests that the host specificity and host range of P22-like phages may vary. Here, we analyzed the infected host species in sixty-six P22-like phages.

These phages infected closely related γ -Proteobacteria *Enterobacteriaceae* species (*Salmonella enterica*, *Shigella flexneri* and *E. coli*). These phages with homologous tailspike proteins can infect similar host strains of *Salmonella* Typhimurium (Table S3), including LcII-101, LcI-40 13-am43, BTP1, SPN9CC, S9-5, SE16, SE21, vB_SenP_ER25, P22, S149, ST104, ST160, ST64T, vB_SaIP_ABTLNsp11242, vB_SaIP_PM43, MG40, ST-29, ST-32, ST-87 and ST-35, whereas phages CUS-3, Sf6, vB_PmiP_Brookers and epsilon34 with different tailspike proteins can infect

E. coli, *Shigella flexneri*, *Proteus mirabilis* and *Salmonella* Anatum, respectively. In this study, the amino acid sequence of the tailspike protein was 93% identical to that of HK620. The HK620 tailspike possesses endo-N-acetylglucosaminidase activity and creates hexasaccharides of an O18A1-type O-antigen, according to a previous study (Barbirz et al., 2008). Phage E212 and HK620 could recognize and cleave *E. coli* with homologous tailspike proteins.

It is known that the ATPase and endonuclease activities of terminase proteins allow dsDNA molecules to be bundled into a procapsid and cleaved into genome-sized lengths. These proteins, including the portal protein, terminase, and major capsid protein, play an important role in DNA packaging. The terminases of tailed phages consist of a large and a small subunit. The large subunit possesses an N-terminal domain that hydrolyzes ATP and a C-terminal domain that possesses nuclease activity and DNA translocation function (Sun et al., 2008). The small subunit of the phage P22 genome encapsulates DNA by recognizing a specific DNA sequence known as the *pac* site (Wu et al., 2002). The DNA packaging modules of phage E212 had two terminase subunits with significant amino acid sequence identity (82% to *Salmonella* phage 118, 970_sal4 and 94% to *Enterobacteria* phage IME10). The DNA packing tactics of tailed dsDNA phages can be categorized into six kinds and seventeen subtypes: (a) cohesive ends (5' cos, lambda P2; 3' cos, HK97); (b) headful packaging (P2, P22, Sf6, T4, 933 W, phiPLPE, phiKZ); (c) host ends (Mu, D3112); (d) short direct terminal repeats (DTRs) (T7, N4, C-st); (e) long DTRs (SPO1); and (f) covalently bound terminal proteins (*Bacillus subtilis* phage phi29) (Casjens et al., 2009; Merrill et al., 2016). It is possible to predict the packaging strategy of a tailed phage from the amino acid sequence of its major terminase subunit. As shown in Fig. 6b, the major terminase subunit of E212 shared a significant degree of identity with other *Lederbergvirus* members. These phages with terminally redundant and circularly permuted chromosomes undergo a process known as "headful packaging" (Streisinger et al., 1967). In this packaging process, the initiation site is designated *pac*, and the terminase performs sequence-specific cleavage to commence the process. However, the terminase has limited sequence specificity at the subsequent headful cleavages, which are only made when the procapsid is filled with DNA. Typically, the packed DNA length in headful packaging phages is between 102% and 110% of the length of the genome sequence; hence, these chromosomes feature direct terminal repeats that range from 2% to 10% of the genome length in different phages (Casjens et al., 2009). In a study by Casjens et al., the length of the P22 chromosome was 43,400 base pairs, the length of the genome was 41,830 base pairs, and the terminal redundancy was 1600 base pairs, or 3.8% (Casjens et al., 1988). The DNA ends of headful packaging phages are typically believed to be blunt because they produce blunt ends (Schmiegner et al., 1990).

Depending on the condition of the host cell, lysogeny control and antitermination determine the lytic/lysogenic cycles of phages (Shin et al., 2014). The temperate phage can either enter the lytic cycle, producing progeny virions upon lysis of the host cell, or lysogenize the host cell by integrating its genome as a prophage into the host genome (Liang et al., 2019). Lysogeny is normally quite stable when maintained by sufficient quantities of phage repressors (ORF 9, 10, and 20); however, the lysis-lysogeny switch may have a substantial impact on host metabolism, population dynamics, ecological processes, and phage propagation (Oz et al., 2017). The transcription-activator protein CI of phage P22 is essential for the lytic/lysogenic transition of the phage. In a study by Ho et al., it was determined that P22 CI is a tetrameric protein made of four subunits, and there are substantial similarities between CII and P22 CI in terms of their sequence, size, and function, as well as their recognition sequence (TTGC—N6-TTGC/T) (Ho et al., 1992). In this study, the amino acid sequence of the E212 CI protein and CII protein were 0% identical to that of P22 CI and CII, but the amino acid sequence of the E212 CI protein was 37% identical to P22 CII protein. The liberation of progeny from the host cell wall is an essential step for successful competition in the infection cycle. Most dsDNA bacteriophages depend

on the holin-endolysin system, and holin forms lesions in the cytoplasmic membrane that allow endolysin to reach the murein layer. In general, these two proteins are sufficient to lyse cells and release progeny. However, gram-negative bacteria, such as *E. coli*, *Salmonella*, and *Shigella*, possess an outer membrane in addition to the cytoplasmic membrane and peptidoglycan. To break the outer membrane, many dsDNA bacteriophages encode additional lysis proteins (Summer et al., 2007). It is believed that the Rz protein possesses endopeptidase activity, which may be implicated in the breakdown of oligopeptide crosslinks between glycosidic strands in the peptidoglycan and the lipoproteins of the outer bacterial membrane. This analysis predicted the presence of three nearby proteins in lysis cassettes: ORF23 was discovered to encode Rz, which degrades the final outer membrane of the host cell, holin (ORF21) forms holes in the cytoplasmic membrane, and lysin (ORF22) degrades the cell wall peptidoglycan layer.

5. Conclusion

In this study, the whole nucleotide and amino acid sequences of vB_EcoP_E212 were effectively examined. It is a temperate bacteriophage with 38,252 bp of linear double-stranded DNA with a total G + C content of 46.98%. According to genome analysis and morphological characteristics, E212 belongs to a novel genus, *Lederbergvirus*. Phages are an ideal candidate for replacing antibiotics, but not all phages can be considered for this approach. Phages that carry genes for lysogens, antibiotics, and virulence should not be used as biocontrol agents. The results of this study enrich our understanding of phages and their diversity.

CRedit authorship contribution statement

Bingdong Wei: Methodology, Formal analysis, Writing – original draft. **Cong Cong:** Methodology, Writing – review & editing. **Lin Zheng:** Writing – review & editing. **Long Chen:** Writing – review & editing. **Xiaogang Yan:** Conceptualization, Resources, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

Nucleotide sequence accession number

The GenBank accession number for phage vB_EcoP_E212 is MZ043897.1

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Ethical approval

This article does not contain any studies with human participants or animals by any of the authors.

Supplementary materials

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